

Sink or Swim: Planning for Climate Change



Image Source: The Organic Startup Blog (5 Ways to Implement Adaptability in Your Business – Origins of Species Ch 4)
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Atlantic Association of Planning Technicians

To promote and advance the interests, education and experience of the Planning Technician

2016 Workshop Program

 Centre of Geographic Sciences
COGS | NSCC

June 13-14-15, 2016

Website

www.aapt.ca

Contact

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Special Guest Presenter:

Richard Zurawski BSc MA (Research)
PhD Candidate

as seen and heard on:



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Welcome

AAPT President's Message

On behalf of the Atlantic Association of Planning Technicians, it is my great pleasure to welcome you to the AAPT Workshop for 2016. This year's workshop will carry the theme of Climate Change and will consist of a variety of boardroom lectures, hands-on lab sessions, and field trips.

For those of you that may be new to this association and/or attending our workshop for the first time, I encourage you to network with the other workshop attendees and presenters as much as possible and trust that you will find this workshop educationally beneficial and rewarding.

I would like to thank the NSCC Centre of Geographic Sciences staff and faculty for hosting the workshop, the presenters, our facilitators, the attendees, the organizers, and our sponsor; Nova Scotia Department of Municipal Affairs. These workshops would not be possible without each of you.

Best Regards,



Scott LeBlanc, AAPT President



About the cover

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2016 AAPT Workshop Agenda

Sink or Swim: Planning for Climate Change



Day One—Monday June 13, 2016

8:30—9:00 am	Welcome – coffee, refreshments, snacks	boardroom
9:00—10:00	Keynote Speaker James Boxall, Prof. Marine Affairs and Geography - Dalhousie GIScience Centre	boardroom
10:00—12:00	UAV (Drone) Mapping Sean Fraser, Geomatics Sales Agent - SDF Solutions	boardroom and field trip
12:00—1:30 pm	Lunch	
1:30—3:00 pm	The HRM Urban Forest Master Plan (UFMP) John Charles, Planner - Halifax Regional Municipality	boardroom
3:00—5:00 pm (starts at 3:30pm)	Tour of Annapolis Tidal Power Plant 236 Prince Albert Road, Annapolis Royal	field trip

Day Two—Tuesday June 14, 2016

8:30—9:00 am	Welcome – coffee, refreshments, snacks	boardroom
9:00—10:30	Media and Climate Change Richard Zurawski, BSc MA (Research) PhD Candidate	boardroom
10:30—12:00	Principles and Methods for Visualization of Sea Level Rise and Flooding Eric Rapaport PhD, Associate Professor - School of Planning, Dalhousie University	boardroom
12:00—1:30 pm	Lunch	
1:30—3:00 pm	Sea Level Rise Modelling Lab Scott LeBlanc, Planning Technician II - NS Department of Natural Resources	Lab 152
3:00—3:30 pm	Review of UAV (Drone) Mapping - Collected Data Scott LeBlanc, Planning Technician II - NS Department of Natural Resources	Lab 152
3:30—5:00 pm	Annual General Meeting	boardroom

Day Three—Wednesday June 15, 2016

8:30—9:00 am	Welcome – coffee, refreshments, snacks	boardroom
9:00—12:00	Examining Climate Change GIS Data in ArcGIS Ed Symons, Geomatics Faculty - NSCC Centre of Geographic Sciences	Lab 152
12:00 pm	Workshop Closing Remarks	boardroom

Note: there will be morning and afternoon breaks each day as determined by the presenters and workshop facilitators

Mini-Abstracts

UAV (Drone) Mapping

Sean Fraser, Geomatics Sales Agent (SDF Solutions)

Drone Service
Providers in North
America



(Infographic: Anya
Lamb, Marketing
Manager)

From a delivery time standpoint, a drone allows you to take to the skies virtually whenever you like – collecting the geo-referenced images you require without the hassle of booking and waiting for manned aircraft or satellite imagery. In terms of resolution, systems such as the eBee can achieve orthomosaic/3D model accuracy of roughly 1-3x GSD, or down to 4.5 cm (1.8 in) per pixel – significantly higher than current satellite and manned aircraft can attain. (Source: <https://www.sensefly.com/applications/gis.html>)

This session will provide you with an equipment overview, a review of a pre-flight checklist, an overview of the establishment of ground control points, and finishes off with a field demonstration including data collection.

The HRM Urban Forest Master Plan (UFMP)

John Charles, Senior Planner (Halifax Regional Municipality)

The presentation will focus on the development of the HRM Urban Forest Master Plan (UFMP) and the role that GIS has played in the canopy model in UFMP research and policy development in HRM's urban forest. In the management of a city's urban forest, the levels of planning range from strategic, city-wide management planning to detailed, street level planning. It is the vision of the HRM UFMP to overcome disconnects between spatial levels of management and gaps between policy and operations. This has been accomplished through the neighbourhood management approach that includes four discrete spatial levels, from broadest to finest: 1) the UFMP Study Area, which essentially includes all municipally serviced areas in HRM, 2) the Communities, which follow boundaries of historical pre-amalgamation cities and towns, 3) the Neighbourhoods, which exhibit distinctive environmental and settlement patterns, and 4) the Neighbourhood Divisions, featuring unique land-use subsets within neighborhoods that warrant special attention in local urban forest management.

Tour of Annapolis Tidal Power Plant

236 Prince Albert Road, Annapolis Royal

Nova Scotia Power operates the first and only tidal power plant in North America, one of few in the world. The Annapolis Tidal Power Plant came online in 1984. It can make as much as 20 megawatts of electricity and has a daily output of roughly 80-100 megawatt hours, depending on the tides. The tour will begin with NSPI Plant crew and then handed over to the Visitor Information Centre staff who will take us on a unique "Down Under" tour of the tidal power plant. We will travel by elevator below ground to the upper machine room, directly above the turbine. This will give you a first-hand look at how this power plant works along with a tour guide to explain it all.

Mini-Abstracts

Media and Climate Change

Richard Zurawski, BSc MA (Research) PhD Candidate

In the sciences we often think that by presenting the 'facts' and the 'evidence' with respect to scientific research, we have done our jobs as scientists. Unfortunately the evidence tell us that this is not quite true. We live in an era of 'hyper-communication' an extension of what Hunter Thompson said was "a long cruel, shallow money trench, where pimps and thieves run free and good men die like dogs.' Todays rampant pseudo-scientific memes promoted by vested interests have made this quote more prescient than even Thompson could imagine. The sad fact is that now, in addition to researching the sciences, scientists must also take on the mantle of 'science communicators for the public' as well. Why? Because there is a lot at stake. Human Induced Climate Change (HICC) is perhaps one of the best examples I can think of that illustrates this very issue to a tee.

In spite of the unerring progress of scientific method based on something called the 'deficit model' of information transference, scientists have been pushed further and further out of the mainstream of science communication. The implication that people make their decisions objectively and are 'fact-based' has been shown to be largely incorrect and doesn't function well in mass media.

By not understanding a very, very simple human trait, Climate Scientists opened the door to vested interests, and missed an opportunity to squelch the anti-climate lobby and decades of inaction and fruitless argument.

Principles and Methods for Visualization of Sea Level Rise and Flooding

Eric Rapaport PhD, Associate Professor (School of Planning, Dalhousie University)

This presentation will cover the principles of good visualization methods based on my experiences as well as international research. These principles will be discussed through the demonstration of different visualization tools and techniques based on case studies in Nova Scotia completed by honor thesis and graduate students.

Sea Level Rise Modelling Lab

Scott LeBlanc, Planning Technician II (NS Department of Natural Resources)

For those of you who are NSCC COGS graduates you will be transported back to your college days with this hands-on self-guided exercise to model the impact of sea level rise in an existing provincial park. You will work with bringing data over to SketchUP from ArcGIS, creating a surface derived from contours, draping features such as waterbodies, parking lots, roads, boardwalks, etc., and testing the effects of sea level rise on the park. The results will be compared to oblique photography of the areas of the park that have experienced inundation from ocean waters over the years to validate the accuracy and usefulness of the modelling.

Review of UAV (Drone) Mapping - Collected Data

Scott LeBlanc, Planning Technician II (NS Department of Natural Resources)

In this lab session we will review the data that was collected during the UAV (Drone) Mapping session from day one of the workshop and discuss the applicability of such data to your profession.

Mini-Abstracts

Examining Climate Change GIS Data in ArcGIS

Ed Symons, Geomatics Faculty (NSCC Centre of Geographic Sciences)

Satellite observation and climate model data are becoming more and more widely used in GIS. Network Common Data Form (netCDF) is a file format for storing multi-dimensional scientific data (variables) such as temperature, humidity, pressure, wind speed, and direction. NetCDF format is not a traditionally used GIS format although it is getting popular in the community. This session will allow participants to import data in NetCDF format into ArcGIS. We will investigate using the Multi-dimension Tools toolbox to create a raster layer, a feature layer, and a table view from a netCDF file. Methods for data display and some simple analysis using data derived from a netCDF file will be explored.

Presenter Biographies

James Boxall FRCGS, FRGS, Prof. Marine Affairs and Geography (Dalhousie GIScience Centre)



(picture: B. Bottomley,
Dalhousie University)

James Boxall FRCGS, FRGS is a Governor of the Royal Canadian Geographical Society and Fellow of the Royal Geographical Society, as well as a Governor of the Nova Scotia Museums. Over three decades of teaching, he has led the creation and operation of the GIScience Centre at Dalhousie where he teaches geography, GIS, and Marine Spatial Planning. His research focuses on the use of geospatial data and GIS to enhance geoliteracy, with a particular emphasis on youth understanding of oceans and climate change. He has received the Canadian Geography Education Medal, and the Erebus Medal for the Franklin Expedition, as well as accepting the ESRI Special Achievement Award for GIS in 2012. He is a past chair of the NRCan Pan-Canadian Roundtable on Geomatics (now GeoAlliance Canada) and led the development of the Canadian Geographic Education Council. He is a past-President of the Canadian Cartographic Association and the Geomatics Association of Nova Scotia, and is currently the regional representative for the International Geographical Union Commission on Geographic Education.

Sean Fraser, Geomatics Sales Agent (SDF Solutions)

Sean has 15 years of industry/sales experience in the Geomatics, Forestry and Positioning industry. Graduating from the Maritime College of Forest Technology in 2000 Sean began his career working for the Department of Natural Resources as a Forest Technologist before attending the NSCC Centre of Geographic Sciences to get his Advanced Diploma in Geographic Information Systems while specializing in Remote Sensing. Since then Sean has been all around the world including the Turks and Caicos, US, and Canada selling, supporting and utilizing the latest in positioning technology from world leading manufacturers. Sean is a passionate father and family man and can be found at the hockey rink on the weekends or travelling to his camp for a fishing/hunting trip.

John Charles, Planner (Halifax Regional Municipality)

John Charles is a Planner with Halifax Regional Municipality (HRM). Since 2010, he has served as project manager for the HRM Urban Forest Master Plan (UFMP). During this time, John has worked closely with John Simmons, HRM's Urban Forester, as well as Dr. Peter Duinker and his dedicated team of graduate and post graduate students at Dalhousie University's School for Resource and Environmental Studies. The UFMP was adopted by HRM's Regional Council in September, 2012. Since then, the UFMP project team has focused on Plan implementation. To date, the City has planted over 4,500 new street trees and has developed a trial collaborative proactive pruning program with Nova Scotia Power. In addition, HRM has partnered with volunteers throughout the region to plant over 12,000 seedlings and hardwood whips in City parks. In 2014, the UFMP gained national recognition for excellence in planning. HRM and Dalhousie University received the Canadian Institute of Planners Merit Award for New and Emerging Planning.

Presenter Biographies

Richard Zurawski, BSc MA (Research) PhD Candidate



(picture: <http://www.richardzurawski.ca>)

Richard is an educator and a life long learner with a specialization in the education and communications as it relates to the sciences. Richard has a B.Sc. in Physics, an M.A. in Educational Research and is also a PhD Candidate. He is a part time faculty member in the department of Engineering at SMU teaching engineers to write and communicate. His public outreach includes being a documentary film maker, keynote speaker, an author with four published books, on-air meteorologist and a radio talk show host and a business entrepreneur.

His career in the media has covered virtually all media (print, radio, television, and new media) in almost every capacity, from journalist to host to writer to producer and illustrates my communications skills in identifying, developing and negotiating innovative technologies, partnerships, strategic alliances and relationships with support from private and public sectors. As a popularizer of the sciences, he is unique.

Most significantly within the Rogers Radio Broadcast Group umbrella he hosts on Saturday mornings (11-12) a one-hour, all science, call-in, radio talk show on News95-7, "Richard Zurawski's Science Files". He also cohosts the regular weekly broadcasts of the "Science Files" on the Rick Show in Halifax and the Eric Drodz Show in Kitchener (570News) as well as being the on-air meteorologist and science authority for the Rogers broadcast group in Alberta, Ontario and the Maritimes.

His work to promote science and learning on all levels was rewarded in November of 2013 with the Discovery Centre's Science Champion Award.

Eric Rapaport PhD, Associate Professor (School of Planning, Dalhousie University)

Eric Rapaport is a professor at the School of Planning, Dalhousie University. He teaches geographical information systems and planning methods course to both undergraduate and graduate courses. His research focuses on the interaction of the natural systems and the build environment. He has been working on research over the past years on different methods to visualize sea level rise with undergraduate and graduate students.

Scott LeBlanc, Planning Technician II (Regional Services Branch - DNR Nova Scotia)

Scott LeBlanc is a graduate of the Planning/Land Information Technology Diploma Program from the College of Geographic Sciences and has been working in the Geomatics industry for the past 20 years with consulting/engineering firms, local governments, and the Provincial government. He is currently working with the Parks, Outreach, and Service Delivery Division, in the Regional Services Branch at the Department of Natural Resources, Nova Scotia. Scott has been a member of the Atlantic Association of Planning Technicians since 2005 and has been the President of the association since 2010. Scott lives in North River, NS with his beautiful wife and AAPT Treasurer, Tonia, and their energetic son, Evan.

Presenter Biographies

Ed Symons, Geomatics Faculty (NSCC Centre of Geographic Sciences)

Upon first arriving to Nova Scotia in 1995, Ed Symons feels very fortunate to call the Annapolis Valley his home. Since 2001, Ed has been a member of the Geomatics Faculty at COGS and has facilitated a number of courses in mapping, GIS and planning. His interests in community collaboration, open data and wrestling with the concept of resilience have led him to forge an innovative partnership with local community members keen on mapping. Ed received his Bachelor of Science at Lakehead University in Thunder Bay, Ontario and a Diploma in Planning: Land Information Technology at COGS.



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